

The authaffil package

First-appearance numbering of author affiliations

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Abstract

`authaffil` removes hand-numbered affiliation superscripts from multi-author papers. You define each affiliation once under a short code, tag each author with codes, and the package assigns and prints the numbers in the order the codes are *first cited*. Adding, removing, or reordering authors rennumbers everything automatically.

1 Installation

Place `authaffil.sty` next to your `.tex` file (or anywhere on your $\text{\TeX}$ path) and load it:

```
\usepackage{authaffil}           % default: per-author numbers sorted ascending
\usepackage[nosort]{authaffil}   % per-author numbers in citation order
```

It requires only `expl3` (present in every modern $\text{\TeX}$ distribution) and works with $\text{pdf\LaTeX}$, $\text{Xe\LaTeX}$, and $\text{Lua\LaTeX}$.

2 Quick start

```
% 1. Define affiliations (the order of these lines is irrelevant).
\DefineAffiliation{HMS}{Dept. of Biomedical Informatics, HMS, Boston, MA, USA}
\DefineAffiliation{OXE}{Dept. of Engineering Science, University of Oxford, UK}
\DefineAffiliation{ETH}{Dept. of Computer Science, ETH Zurich, Switzerland}

% 2. List authors, tagging each with codes (+ optional footnote symbols).
\AddAuthor{Ayush~Noori}{HMS,OXE}
\AddAuthor[*]{Manuel~Burger}{ETH}
\AddAuthor[\dagger,\ddagger]{Marinka~Zitnik}{HMS}

% 3. (Optional) legend lines.
\AddAffilNote[*]{Equal contribution.}
\AddAffilNote{\dagger}{Jointly supervised this work.}

% 4. Typeset -- e.g. inside \author{...} of your template.
\author{{\small\begin{center}
  \PrintAuthors      \\\[2mm]
  \PrintAffiliations \\\[1mm]
  \PrintAffilNotes
\end{center}}}
```

The block above renders as:

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¹Dept. of Biomedical Informatics, HMS, Boston, MA, USA

²Dept. of Engineering Science, University of Oxford, UK

³Dept. of Computer Science, ETH Zurich, Switzerland

*Equal contribution.

†Jointly supervised this work.

3 Command reference

`\DefineAffiliation{<code>}{<address>}` Register one affiliation. `<code>` is any short string—three-letter codes are convenient but not required. `<address>` may contain `\\` for forced line breaks. Call once per affiliation, in any order.

`\AddAuthor[<symbols>]{<name>}{<codes>}` Append one author. `<codes>` is a comma-separated list of affiliation codes; each new code receives the next number the first time it is seen. The optional `<symbols>` argument is extra superscript material (e.g. `*`, `\dagger`, `\ddagger`) appended after the numbers—write it pre-separated the way you want it to read, e.g. `[\dagger,\ddagger]`. Use `~` inside names to keep them from breaking across lines.

`\AddAffilNote{<symbol>}{<text>}` Register one legend line, typeset as `<symbol><text>`. Lines print in the order added.

`\PrintAuthors` Typeset author names, comma-separated, each with its computed superscript. Sets no font size—wrap it yourself, e.g. `{\small \PrintAuthors}`.

`\PrintAffiliations` Typeset the numbered affiliation list, one per line, in first-appearance order. Switches to `\footnotesize`.

`\PrintAffilNotes` Typeset the legend lines, one per line. Switches to `\footnotesize`.

4 Options

`[sort]` (default) Numbers within a single author are shown ascending: 1,5,30, regardless of citation order.

`[nosort]` Numbers within a single author are shown in the order the codes were cited.

5 How the numbering works

A code receives its number the first time *any* author cites it. So for an author who introduces several brand-new affiliations at once, the left-to-right order of that author’s code list fixes those numbers. To force a particular global ordering, control the order of first citation—the simplest lever is the first author who uses each code. Everything downstream (the per-author superscripts and the printed affiliation list) is derived from that, so there are never two places to keep in sync.

6 Diagnostics

The package writes a warning to the log and terminal in two situations:

- unknown-code** An author cites a code that was never defined with `\DefineAffiliation`. A number is still allocated, but no address exists for it.
- unused-affil** A code was defined but never cited by any author, so it does not appear in the list. Checked once, at `\end{document}`.

Neither is fatal; both are there to catch copy-paste slips before submission.

7 Tips

- The commands are layout-agnostic. Drop `\PrintAuthors`, `\PrintAffiliations`, and `\PrintAffilNotes` into whatever `\title/\author` scaffold your journal class expects.
- A trailing `\\` immediately before `\end{center}` raises a spurious “missing `\item`” error in $\text{\LaTeX}$. The print commands therefore use `\\` as a *separator*, never a terminator, so you can place them safely at the end of a centered block.
- If a legend line uses `\href`, load `hyperref` in your preamble; the package does not load it for you.